

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Regular & Supplementary Winter Examination-2023**

**Course: B. Tech.**

**Branch: Mechanical & Allied Engineering**

**Semester: V**

**Subject Code & Name: BTAPE504D & Automobile Engineering**

**Max Marks: 60**

**Date:08-01-24**

**Duration: 3 Hr.**

***Instructions to the Students:***

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

|                                                                                                                               | (CO) | Marks |
|-------------------------------------------------------------------------------------------------------------------------------|------|-------|
| <b>Q. 1 Solve Any Two of the following.</b>                                                                                   |      | 12    |
| A) Define automobile and mention the detail classification of it with example.                                                | CO1  | 6     |
| B) What is a chassis? Explain with the diagram front engine rear wheel drive chassis layout?                                  | CO2  | 6     |
| C) Enlist the main components of an automobile and describe them in detail?                                                   | CO2  | 6     |
| <b>Q.2 Solve Any Two of the following.</b>                                                                                    |      | 12    |
| A) I) What are the Functions of steering system?                                                                              | CO1  | 3     |
| II) Differentiate clearly between the functions of spring and shock absorber of a suspension system.                          |      | 3     |
| B) Explain with the help of diagram the steering linkage of independent suspension system?                                    | CO2  | 6     |
| C) List out the common trouble experienced in automobile suspension system and suggest the remedies in each case.             | CO4  | 6     |
| <b>Q. 3 Solve Any Two of the following.</b>                                                                                   |      | 12    |
| A) Describe briefly with a neat sketch the construction and working of a single plate clutch.                                 | CO3  | 6     |
| B) Explain with a neat sketch the construction and working of a sliding mesh gearbox along with advantages and disadvantages? | CO2  | 6     |
| C) Explain with a neat sketch the construction and working of a differential?                                                 | CO3  | 6     |
| <b>Q.4 Solve Any Two of the following.</b>                                                                                    |      | 12    |
| A) What are the functions of brake? Classify it in detail with examples?                                                      | CO1  | 6     |
| B) Draw a cross-section of automobile tyre and describe the basic constructional features of it?                              | CO2  | 6     |

C) With the help of neat sketch describe the operation of drum brake? CO3 6

**Q. 5 Solve Any Two of the following.** 12

A) Describe with neat sketch the construction and working of lead-acid battery used in automobile? CO3 6

B) Name the pollutants emitted by the IC engines in the atmosphere and mention its effects? CO5 6

C) Explain with a neat sketch the construction and working of magneto-coil ignition system in automobile? CO2 6

**\*\*\* End \*\*\***